

# Assignment 8

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## Problem Statement

To apply WordNet-based Word Sense Disambiguation (WSD) to improve meaning interpretation in ambiguous sentences.

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## Objective

The objective of this assignment is to understand how ambiguous words can have multiple meanings and to implement a method for selecting the correct meaning based on context using WordNet.

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## Introduction

Word Sense Disambiguation (WSD) is a Natural Language Processing technique used to determine the correct meaning of a word in a given context. Many words in English are ambiguous and can have multiple meanings. WSD helps in resolving this ambiguity to improve understanding and accuracy in NLP applications.

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## Terminologies Used

### 1. Word Sense Disambiguation (WSD)

WSD is the process of identifying the correct meaning (sense) of a word based on its context in a sentence.

## 2. Ambiguous Words

Words that have multiple meanings are called ambiguous words. For example, the word "bank" can refer to a financial institution or the side of a river.

## 3. WordNet

WordNet is a lexical database that groups words into sets of synonyms called synsets and provides semantic relationships between them.

## 4. Synset

A synset is a group of words that share the same meaning. Each synset represents a specific sense of a word.

## 5. Lesk Algorithm

The Lesk algorithm is a commonly used method for WSD. It determines the correct meaning of a word by comparing the context of the sentence with the definitions of different senses in WordNet.

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# Methodology

The following steps were performed:

1. Ambiguous sentences containing words with multiple meanings were selected.
  2. The sentences were tokenized into words.
  3. The Lesk algorithm was applied to identify the most appropriate sense of the target word.
  4. WordNet was used to retrieve the definition and examples of the selected sense.
  5. The results were analyzed to verify correct interpretation based on context.
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# Output Description

- **Sentence:** Input sentence containing an ambiguous word.
- **Target Word:** Word whose meaning is to be determined.
- **Sense:** Identified WordNet synset representing the correct meaning.

- **Definition:** Explanation of the selected sense.
  - **Examples:** Example usage of the identified sense.
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## Applications

- Machine translation
  - Information retrieval
  - Chatbots and virtual assistants
  - Text understanding systems
  - Question answering systems
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## Conclusion

The assignment demonstrates how Word Sense Disambiguation can be performed using WordNet and the Lesk algorithm. By analyzing context, the system successfully identifies the correct meaning of ambiguous words, improving the overall understanding of text.

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## Result

Word Sense Disambiguation was successfully implemented using WordNet, and correct meanings of ambiguous words were identified based on context.